

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2 Answer Book

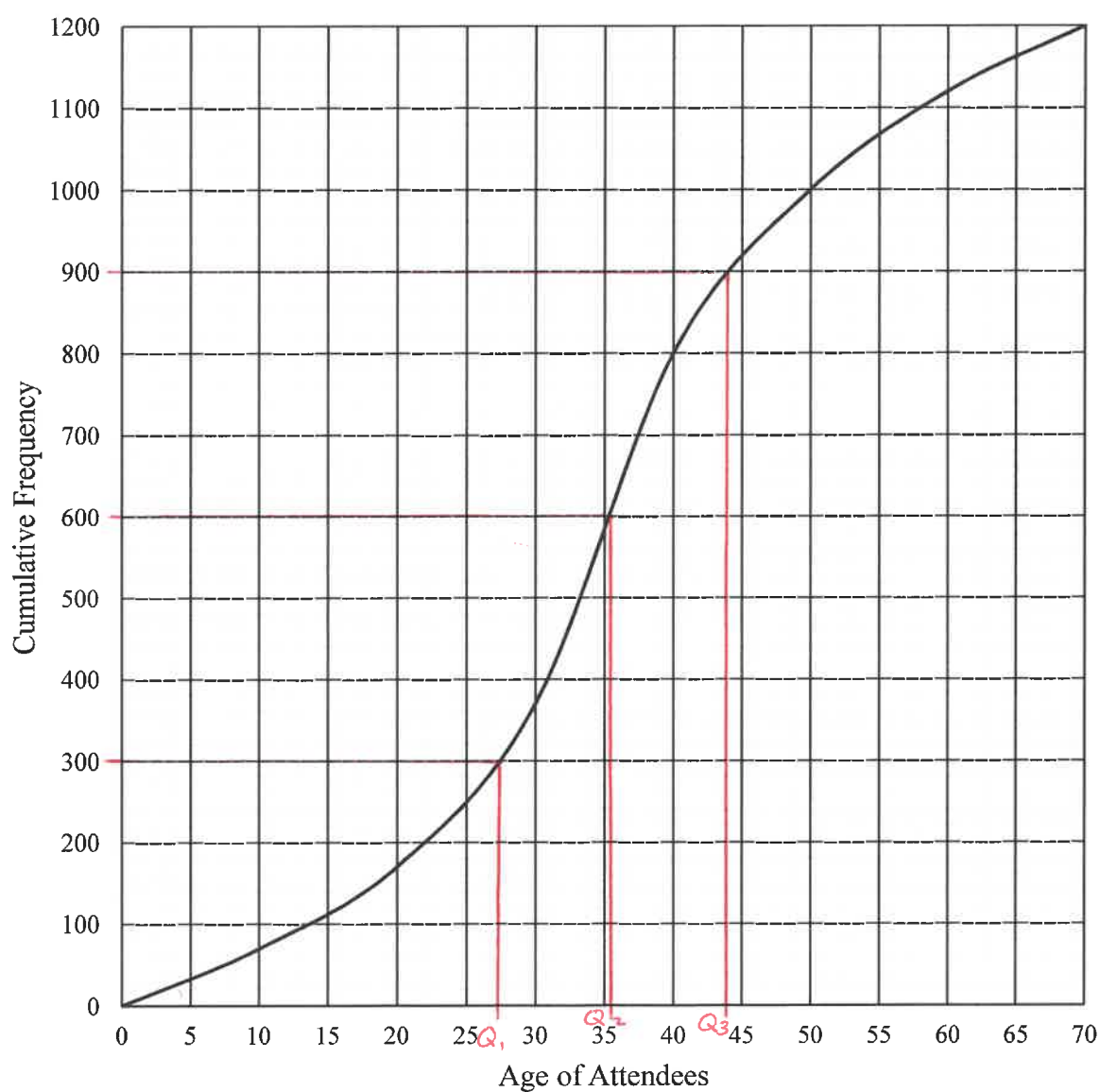
NAME: Memo

CLASS: 12 _____ TEACHER: _____

Question	Mark			Marker	HOD
	H	T	U		
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
TOTAL					
%					

QUESTION 1

Age Group	Number Attended	Cumulative Frequency
$0 < x \leq 10$	70	70
$10 < x \leq 20$	100	170
$20 < x \leq 30$	200	370
$30 < x \leq 40$	A	C
$40 < x \leq 50$	B	D
$50 < x \leq 60$	120	1120
$60 < x \leq 70$	80	1200

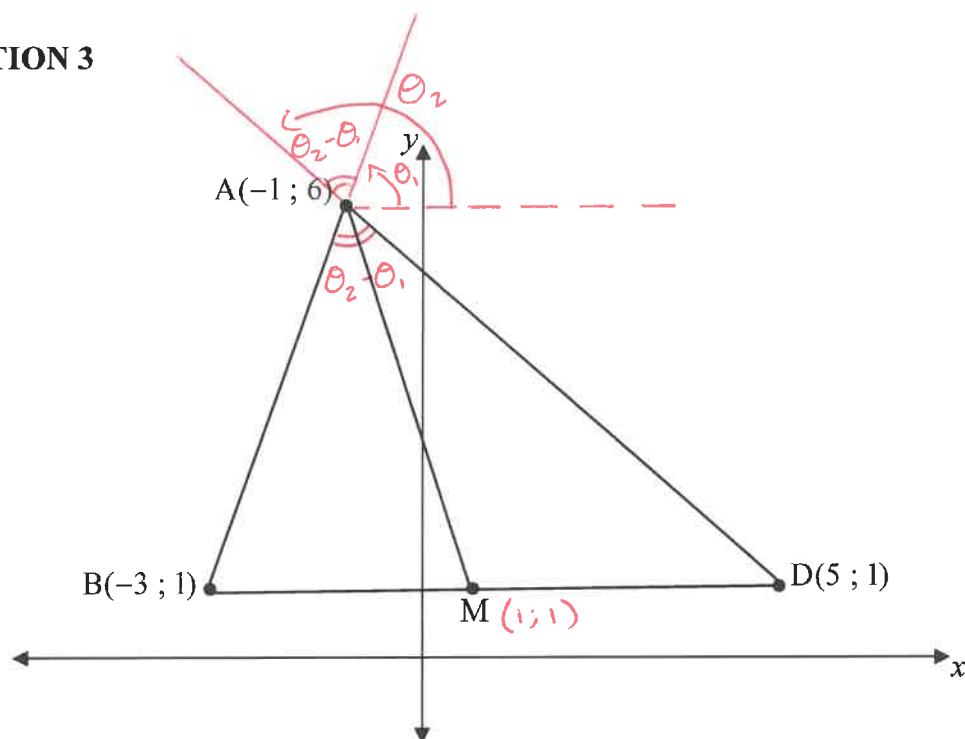


	Solution	Marks
1.1	A = 430 ✓ A B = 200 ✓ A C = 800 ✓ A D = 1000 ✓ A	(4)
1.2	35 years ✓✓ Can accept 36	(2)
1.3	* Can accept ± 1 for Q1, 2 & 3.	(5)
1.4	Symmetrical distribution. ✓ A Box is evenly distributed (not skewed) ✓ Acceptable reason Alt: Ogive has even S shape.	(2)
1.5	It will skew the data to the right / positively skewed. ✓✓ Alt: The data will lie more over the younger ages.	(2)
		[15]

QUESTION 2

	Solution	Marks
2.1	$r = 0,97$ ✓✓	(2)
2.2	The more daily push ups, the stronger the shoulder strength. ✓✓	(2)
Alt:	Strong, positive correlation. ✓✓	
2.3	$\hat{y} = a + bx$	(2)
	$\hat{y} = 9,81 + 0,57x$ ✓ _a ✓ _b	
2.4	$\hat{y} = 9,81 + 0,57(120)$ ✓ _m	(2)
	$y = 78,21$ ✓ _{CA}	
2.5	No. ✓ _A	(2)
	280 push ups a day lies well outside the data set. ✓	
Alt: Mr Clay is much older than the 18 year olds the data was collected from. ✓		[10]

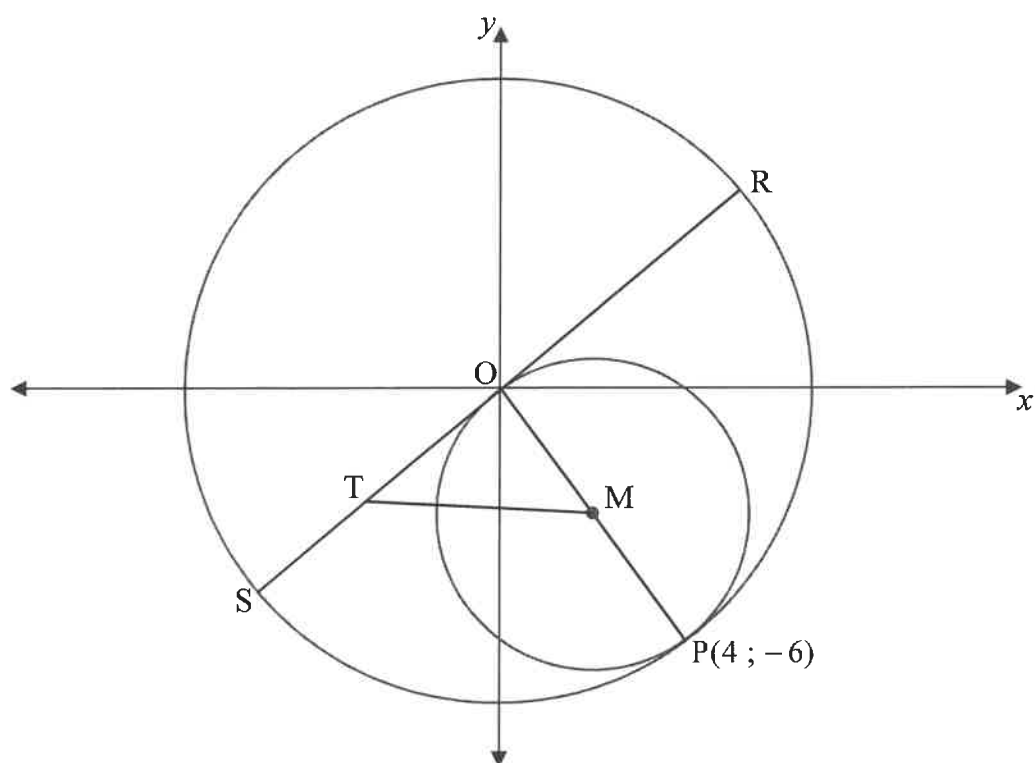
QUESTION 3



	Solution	Marks
3.1	$M_{AD} = \frac{6-1}{-1-5}$ $= -\frac{5}{6}$	(2)
3.2	$M_{BD} = \left(\frac{-3+5}{2}, \frac{1+1}{2} \right) = (1; 1)$ $M_{BD} = M_{AE} \quad \text{S/R Diags of parm}$ $\therefore \frac{x_E - 1}{2} = 1 \quad \frac{y_E + 6}{2} = 1$ $x_E = 3 \quad y_E = -4$ $\therefore E(3; -4) \quad \text{Must be in coordinate form}$	(4)

	Solution	Marks
3.3	$m_{AM} = \frac{6-1}{-1-1} = -\frac{5}{2} \checkmark_{CA}$ $y-1 = -\frac{5}{2}(x-1) \checkmark_m \text{ Alt: Can sub } A(-1;6)$ $y-1 = -\frac{5}{2}x + \frac{5}{2}$ $y = -\frac{5}{2}x + \frac{7}{2} \checkmark_{CA}$	(4)
3.4	$m_{AB} = m_{BC} = m_{AC} \checkmark_m$ $\frac{6-1}{-1+3} = \frac{t-1}{-5+3} \checkmark$ $\frac{5}{2} = \frac{t-1}{-2}$ $-5 = t-1$ $\therefore t = -4 \checkmark_{CA}$	(4)
3.5	$\tan \theta_1 = \frac{5}{2} \checkmark_m \quad \tan \theta_2 = -\frac{5}{6}$ $\theta_1 = 68,2^\circ \checkmark \quad \theta_2 = 140,19^\circ \checkmark$ $\therefore \hat{B\hat{A}D} = \theta_2 - \theta_1 \quad \text{Vert opp } \angle\text{'s}$ $= 71,99^\circ \checkmark$	(4)
		[18]

QUESTION 4



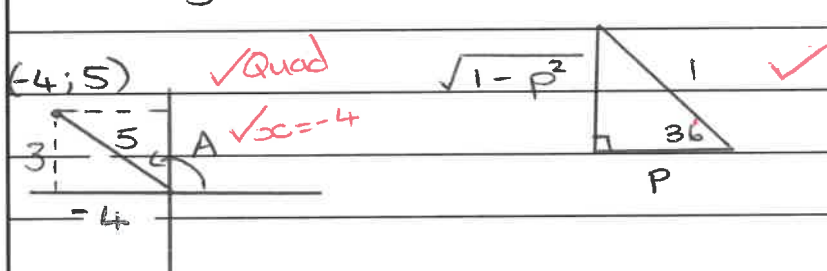
	Solution	Marks
4.1.1	$x^2 + y^2 = r^2$ $(4)^2 + (-6)^2 = r^2$ ✓m $r^2 = 52$ $\therefore x^2 + y^2 = 52$ ✓A	(2)
4.1.2	$M(2; -3)$ ✓ Midpoint of OP $(x-2)^2 + (y+3)^2 = r^2$ ✓CA Equation of C off centre $(4-2)^2 + (-6+3)^2 = r^2$ ✓m subs (4; -6) $\therefore r^2 = 13$ $\therefore (x-2)^2 + (y+3)^2 = 13$ ✓CA	(4)

	Solution	Marks
4.1.3	$m_{OP} = \frac{-6}{4} = -\frac{3}{2} \checkmark$ $\therefore m_{RS} = \frac{2}{3} \quad \text{S/R rad } \perp \text{ tan}$ $\therefore y = \frac{2}{3}x \checkmark$	(3)
4.2	$y = \frac{2}{3}x \quad x^2 + y^2 = 52$ $x^2 + \left(\frac{2}{3}x\right)^2 = 52 \quad \checkmark m \quad RS = \text{Circle}$ $x^2 + \frac{4}{9}x^2 = 52$ $\frac{13}{9}x^2 = 52$ $\therefore x^2 = 36 \quad \checkmark CA$ $x = \pm 6$ $\therefore y = \frac{2}{3}(6) \quad \checkmark m$ $= 4 \quad \checkmark CA \quad \checkmark CA$ $\therefore E(6; 4)$	(5)
4.3	$\hat{O}M = 90^\circ \quad \text{rad } \perp \text{ tan}$ $OT^2 = MT^2 - OM^2 \quad \checkmark m \quad \text{Pythag}$ $OT^2 = (5)^2 - (\sqrt{13})^2 \quad \checkmark A$ $OT^2 = 12$ $\therefore OT = \sqrt{12} = 2\sqrt{3}$ $\approx 3,46 \text{ units} \quad \checkmark \text{Answer accept surd or rounded answer}$	(3)
		[17]

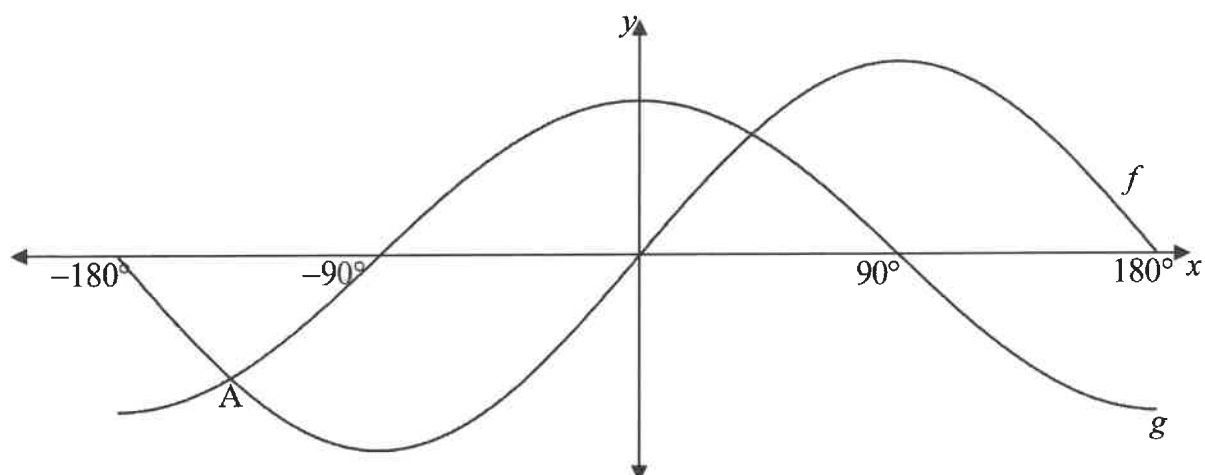
QUESTION 5

	Solution	Marks
5.1.1	$\cos(75^\circ + \theta) \cos(15^\circ + \theta) + \sin(75^\circ + \theta) \sin(15^\circ + \theta)$ $= \cos(75^\circ + \theta - (15^\circ + \theta)) \quad \checkmark \text{compound}$ $= \cos(75^\circ + \theta - 15^\circ - \theta)$ $= \cos 60^\circ \checkmark$ $= \frac{1}{2} \checkmark$	(3)
5.1.2	$\tan(-\theta) \cdot \sin(90^\circ + \theta) + \frac{\sin(360^\circ + 2\theta)}{2 \cos(360^\circ - \theta)}$ $= -\tan\theta \cdot \cos\theta + \frac{\sin 2\theta}{2 \cos\theta} \checkmark$ $= -\frac{\sin\theta}{\cos\theta} \cdot \cos\theta + \frac{2 \sin\theta \cos\theta}{2 \cos\theta} \checkmark$ $= -\sin\theta + \sin\theta \checkmark$ $= 0 \checkmark$	(6)

	Solution	Marks
5.2.1	$\text{LHS} = \frac{\sin 4x \cos 2x - 2 \cos 4x \sin x \cos x}{\tan 2x}$ $= \frac{\sin 4x \cos 2x - \cos 4x \sin 2x}{\tan 2x} \checkmark$ $= \frac{\sin(4x - 2x)}{\tan 2x} \checkmark$ $= \sin 2x \div \tan 2x$ $= \sin 2x \div \frac{\sin 2x}{\cos 2x}$ $= \sin 2x \times \frac{\cos 2x}{\sin 2x} \checkmark$ $= \cos 2x \checkmark$ $= \cos^2 x - \sin^2 x \checkmark = \text{RHS}$ $\therefore \text{LHS} = \text{RHS}$	(5)

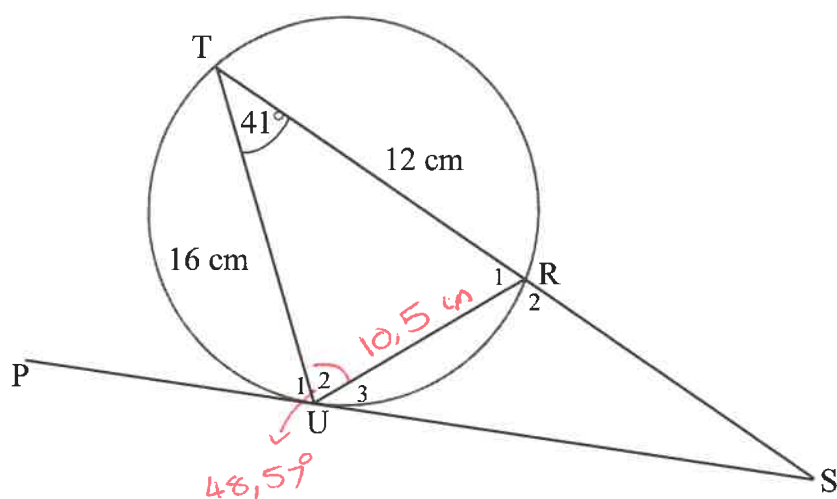
	Solution	Marks
5.2.2	$\tan 2x$ is undefined: $x = 45 + k \cdot 90$; $k \in \mathbb{Z}$. ✓ $\tan 2x = 0$ $x = 0 + k \cdot 90^\circ$; $k \in \mathbb{Z}$ ✓ $\therefore x \in \{0; 45; 90; 135\}$ ✓	(3)
5.3	$\sin^2 x + \cos 2x - \cos x = 0$ $1 - \cos^2 x + 2\cos^2 x - 1 - \cos x = 0$ ✓ $\cos^2 x - \cos x = 0$ ✓ A $\cos x (\cos x - 1) = 0$ $\cos x = 0$ or $\cos x = 1$ ✓ CA both $\therefore x = 90^\circ + k \cdot 180^\circ$; $x = 0 + k \cdot 360^\circ$; $k \in \mathbb{Z}$ ✓ CA $k \in \mathbb{Z}$ ✓ CA	(6)
5.4	$\sin A = \frac{3}{5}$ $\cos 36^\circ = p$  $\cos(-A - 36^\circ)$ $= \cos(A + 36^\circ)$ ✓ $= \cos A \cos 36^\circ - \sin A \sin 36^\circ$ ✓ $= \frac{-4}{5} p - \frac{3}{5} \sqrt{1-p^2}$ ✓ CA ✓ CA	(7)
		[30]

QUESTION 6



	Solution	Marks
6.1	$a = 5$ ✓ $b = 1$ ✓	(2)
6.2	$h(x) = 5 \sin(x - 30) - 3$ ✓ ✓	(2)
6.3	2 ✓ ✓	(2)
6.4	$\frac{5 \sin x}{\cos x} = \frac{4 \cos x}{\cos x}$ $5 \tan x = 4$ $\tan x = \frac{4}{5} \quad \checkmark A$ $\text{ref } L = 38,66^\circ$ $x = 38,66^\circ + k \cdot 180^\circ ; k \in \mathbb{Z} \quad \checkmark \checkmark$ $\therefore x_A = -141,34^\circ \therefore y_A = -3,12$ $\therefore A(-141,34^\circ ; -3,12)$	(5)
		[11]

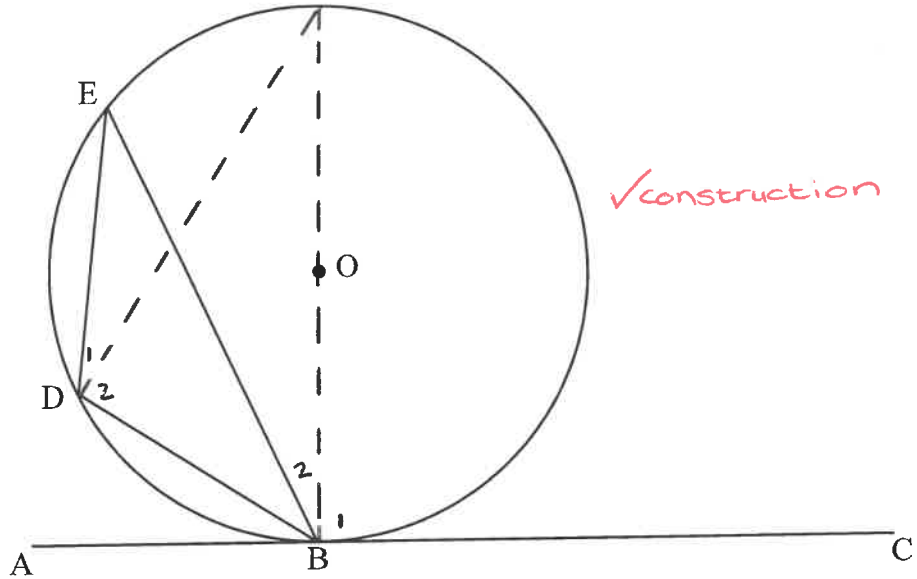
QUESTION 7



	Solution	Marks
7.1	$UR^2 = TU^2 + TR^2 - 2 \cdot TU \cdot TR \cdot \cos \hat{T} \quad \checkmark \text{cos rule}$ $UR^2 = (16)^2 + (12)^2 - 2(16)(12) \cdot \cos 41 \quad \checkmark$ $\therefore UR = 10,50 \text{ cm} \quad \checkmark$	(3)
7.2	$\frac{\sin \hat{U}_2}{12} = \frac{\sin 41}{10,5} \quad \checkmark \text{sin rule}$ $\sin \hat{U}_2 = \frac{12 \cdot \sin 41}{10,5} \quad \checkmark \text{subs}$ $\therefore \hat{U}_2 = 48,57^\circ \quad \checkmark \text{CA depending on UR}$	(3)
7.3	$\hat{S} = 49,43^\circ \quad \checkmark \text{S/R } \angle \text{'s in a } \Delta$ $\frac{TRS}{\sin (89,57)} = \frac{16}{\sin (49,43)} \quad \checkmark \text{sin rule}$ $TRS = 21,06 \text{ cm} \quad \checkmark \text{CA}$	(5)
		[11]

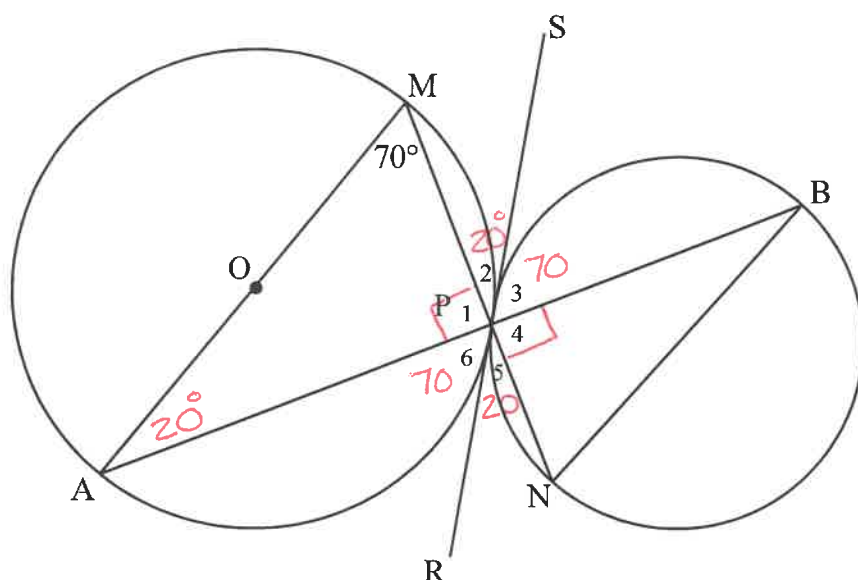


QUESTION 8

	Solution	Marks
8.1	 ✓ construction $\hat{D}_2 = 90^\circ$ S/R $\angle$s in semi $\odot$ $\hat{B}_1 = 90^\circ$ S/R rad $\perp$ tan $\hat{E}DB = 90^\circ + \hat{D}_1$ $\hat{E}BC = 90^\circ + \hat{B}_2$ } ✓ But, $\hat{D}_1 = \hat{B}_2$ S/R $\angle$s in same segment $\therefore \hat{E}DB = \hat{E}BC$ ✓	
		(6)



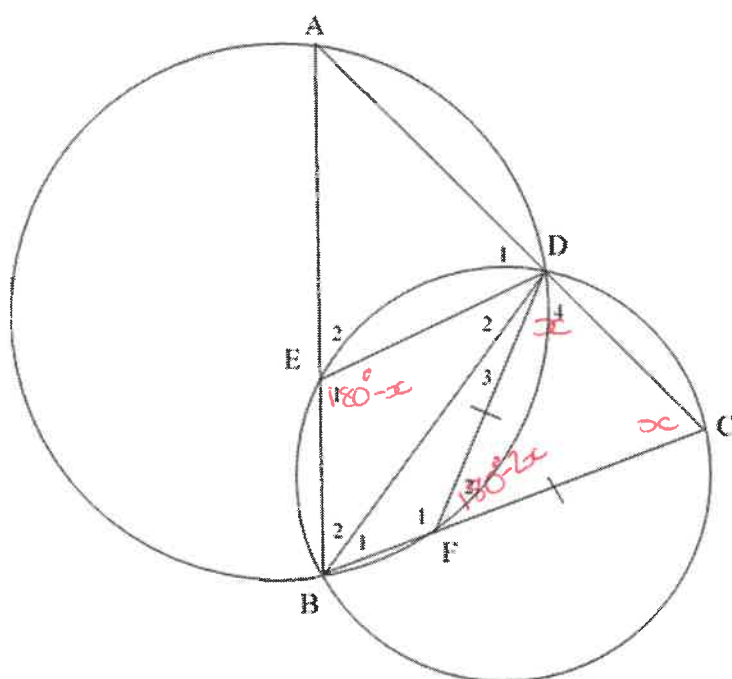
8.2



	Solution	Marks
8.2.1	$\hat{P}_1 = 90^\circ$ ✓ $\angle$'s in semi $\odot$ ✓	(2)
8.2.2	$\hat{P}_1 = \hat{P}_4 = 90^\circ$ $\frac{S}{R}$ Vert opp $\angle$'s $\therefore$ BN is the diameter Converse $\angle$'s in semi $\odot$ ✓ $\frac{S}{R}$ $\frac{Alt}{chord}$ subtends 90°	
8.2.3 (a)	$\hat{A} = 20^\circ$ $\frac{S}{R}$ $\angle$'s in a Δ	(1)
8.2.3 (b)	$\hat{P}_6 = 70^\circ$ ✓ tan chord thrm ✓	(2)
8.2.3 (c)	$\hat{P}_2 = 20^\circ$ $\frac{S}{R}$ tan chord thrm $\hat{P}_2 = \hat{P}_5 = 20^\circ$ $\frac{S}{R}$ Vert opp $\angle$'s $\therefore \hat{B} = \hat{P}_5 = 20^\circ$ $\frac{S}{R}$ tan chord thrm	(3)
		[16]

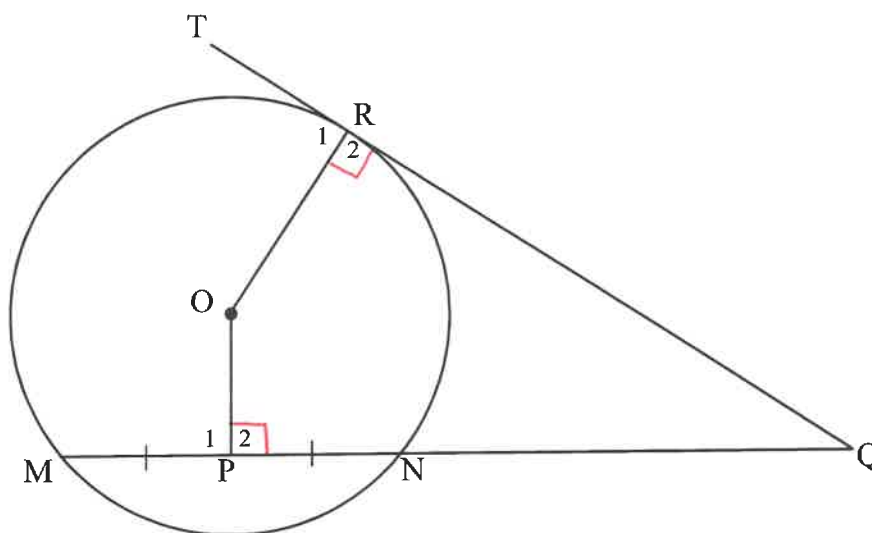


QUESTION 9



	Solution	Marks
9.1.1	$\hat{D}_4 = \hat{C}$ S/R $\angle$'s opp = sides $\hat{C} = x$ S/R $\angle$'s in a Δ $\therefore \hat{E}_1 = 180 - x$ ✓CA Opp $\angle$'s of cyclic quad ✓R	(4)
9.1.2	$\hat{A} = 180 - 2x$ ✓✓ Ext $\angle$ of cyclic quad ✓	(3)
9.2	$\hat{ABF} = x$ S/R $\angle$'s in a Δ $\therefore \hat{E}_1 + \hat{ABF} = 180 - x + x$ $= 180$ ✓ $\therefore ED \parallel BC$ S/R Co-int $\angle$'s supp.	(3)
		[10]

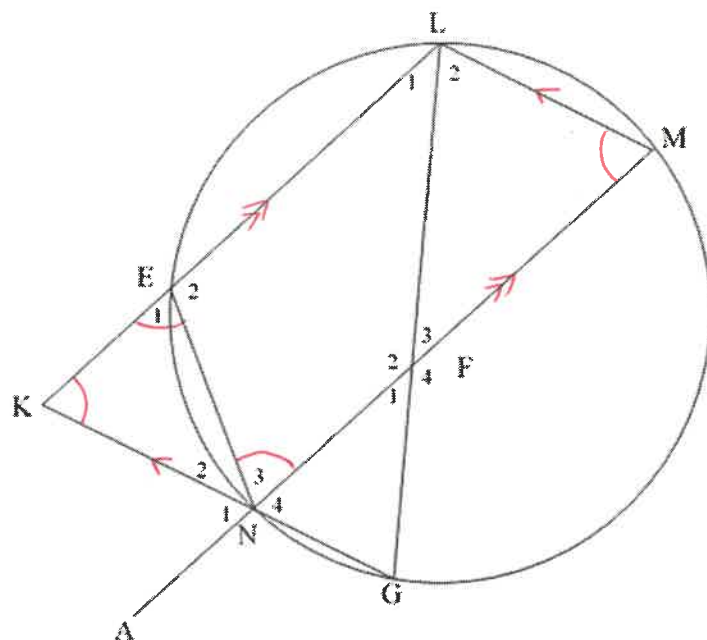
QUESTION 10



	Solution	Marks
10.1	$\hat{P}_2 = 90^\circ$ ✓ line from centre to midpoint chord ✓ $\hat{R}_2 = 90^\circ$ ✓ rad $\perp$ tan ✓ $\therefore \hat{P}_2 + \hat{R}_2 = 180^\circ$ $\therefore OPQR$ is a cyclic quad ✓ converse opp $\angle$'s of cyclic quad Alt: /opp $\angle$'s of quad supp.	(5)



10.2



	Solution	Marks
10.2.1	$\hat{E}_1 = \hat{M}$ S/R Ext $\angle$ of cyclic quad $\hat{K} = \hat{M}$ S/R Opp $\angle$'s of parm $\therefore \hat{E}_1 = \hat{K}$ ✓ $\therefore KN = EN$ S/R Sides opp = $\angle$'s 	(4)
10.2.2	$\hat{E}_1 = \hat{N}_3$ S/R Alt $\angle$'s ; $KL \parallel NM$ $\hat{E}_1 = \hat{K}$ Proven $\therefore \hat{N}_3 = \hat{K}$ ✓ $\therefore MNA$ is a S/R converse tan tangent chord thrm 	(3)
		[12]

